

Overall Transaction Overview

Transaction Overview

User Demographics

Fraud & Security Analysis

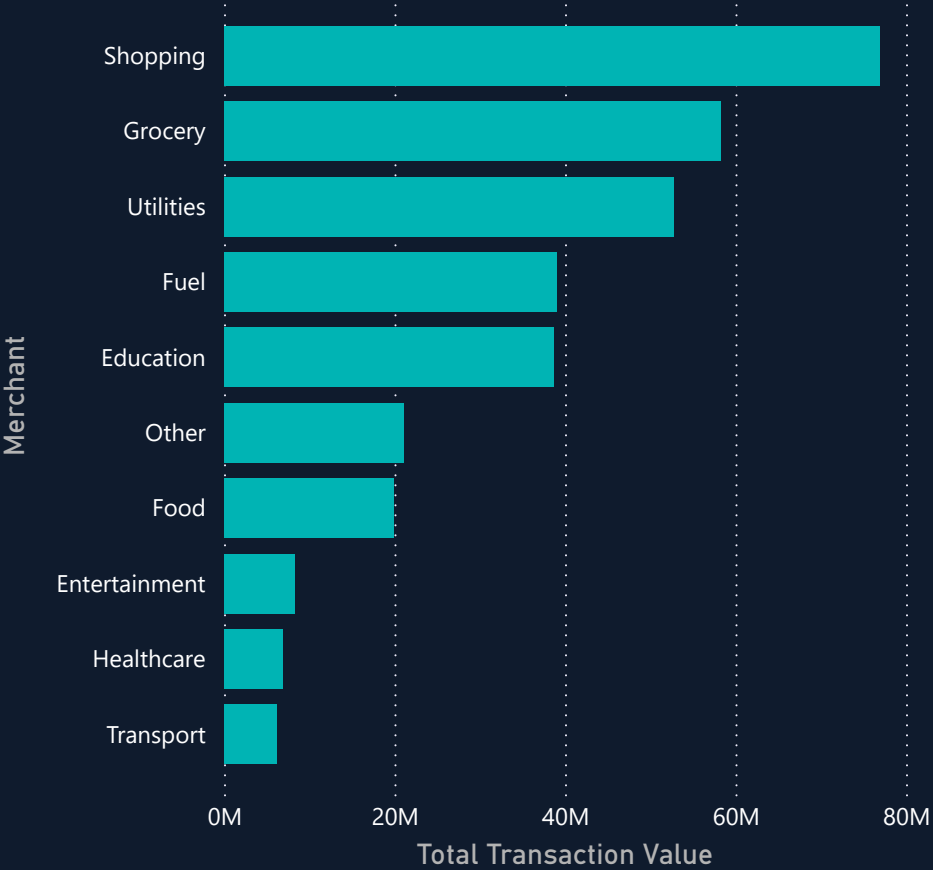
ML Model Performance

Model Evaluation Curves

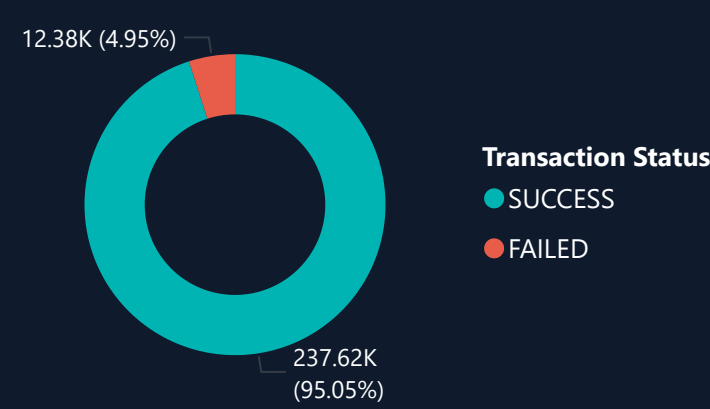
Model Explainability



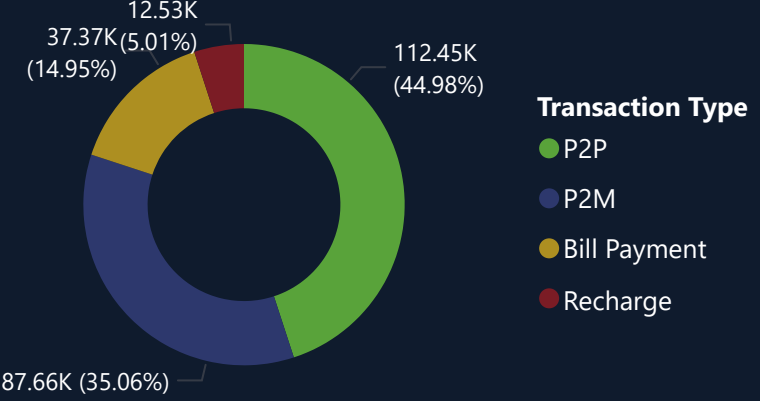
Which Merchants Drive the Most Transaction Value?



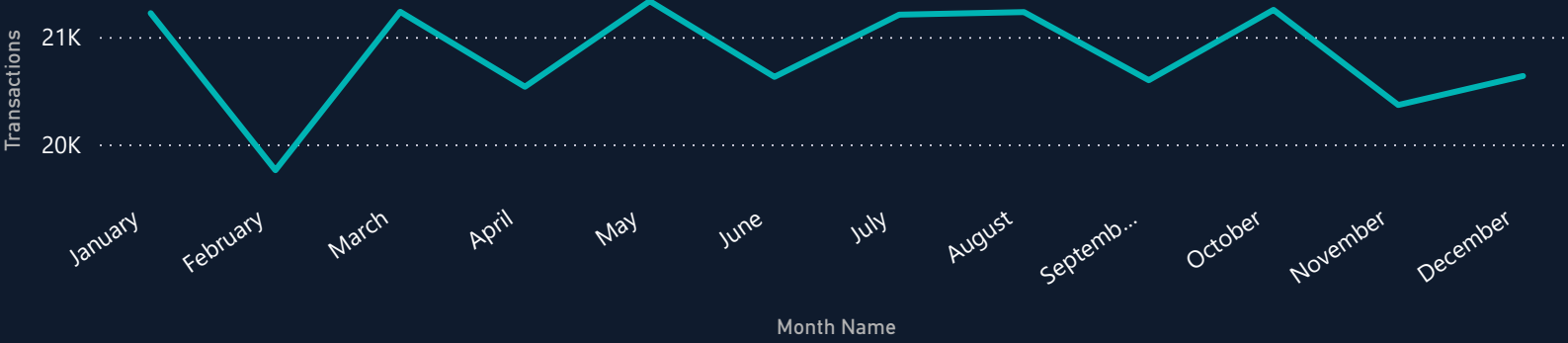
95% of UPI Transactions Complete Successfully



Which Transaction Type Do Users Prefer?



Transaction Volume Stays Consistent Year-Round



User Demographic and Behavior

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Model Explainability

State

All

Age Group

All

250K

Total Transactions

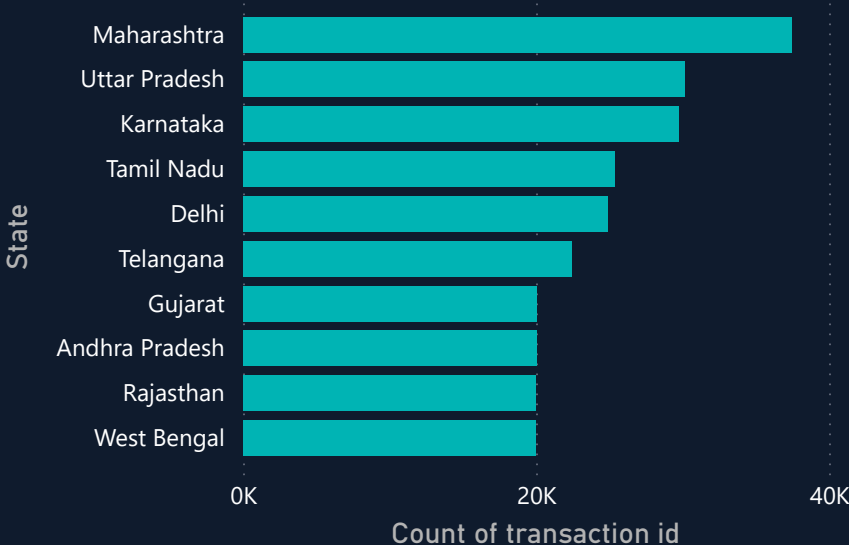
328M

Sum of Amount (INR)

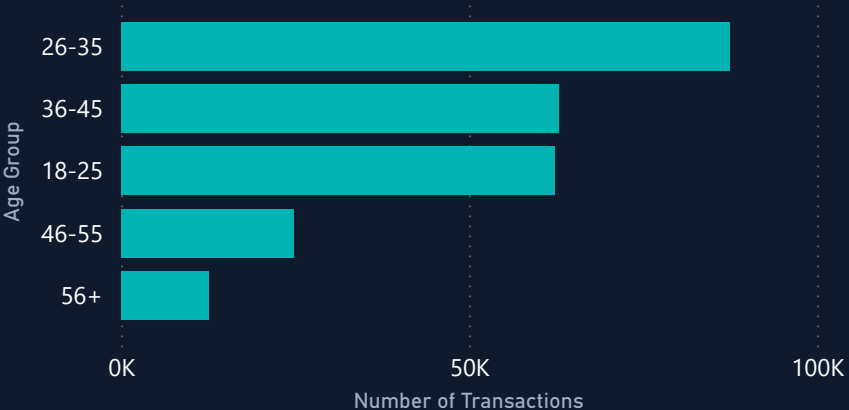
0.95

Success Rate

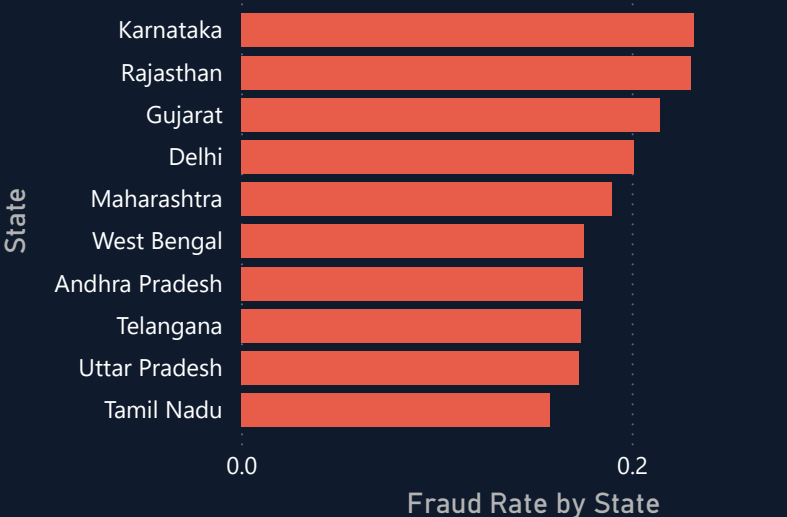
Which States Generate the Most UPI Transactions?



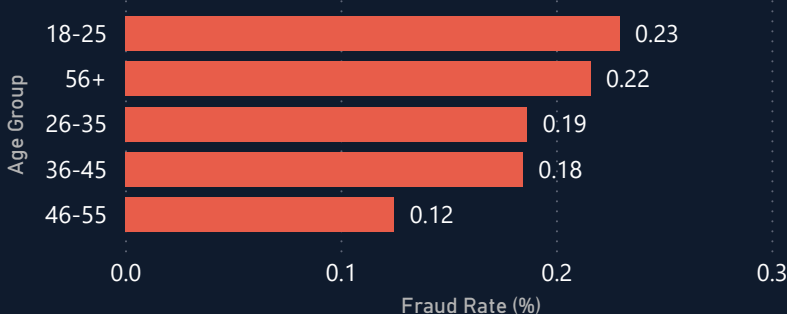
Ages 26–35 Are the Most Active UPI Senders



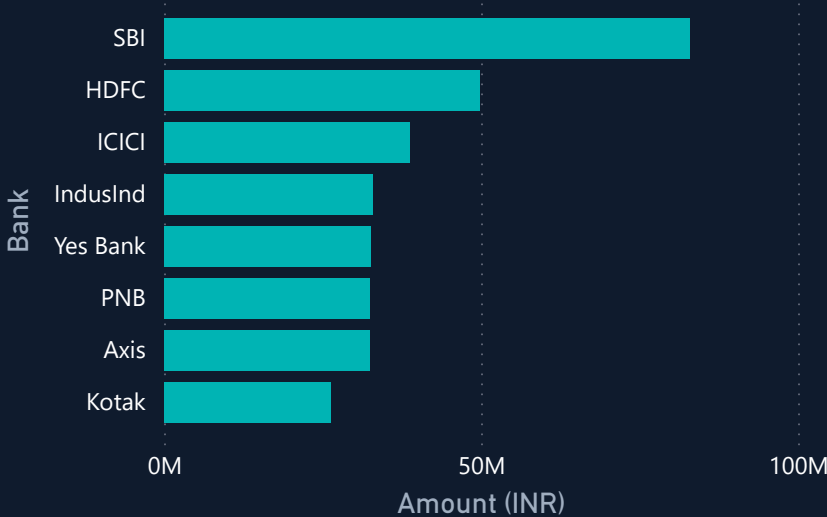
Which States Have the Highest Fraud Rate?



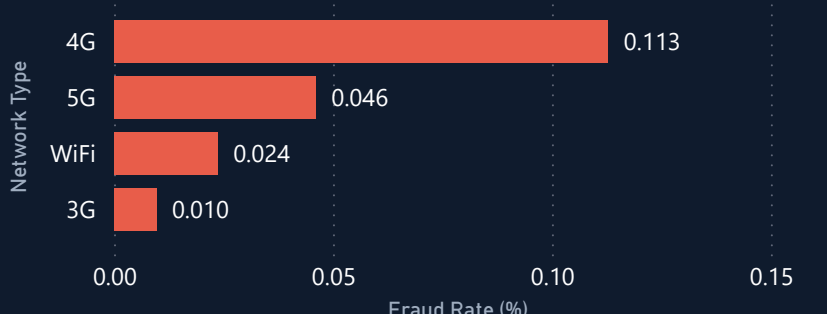
Which Age Group Has the Highest Fraud Rate?



Which Bank Sends the Highest Transaction Value?



Which Network Type Has the Highest Fraud Rate?



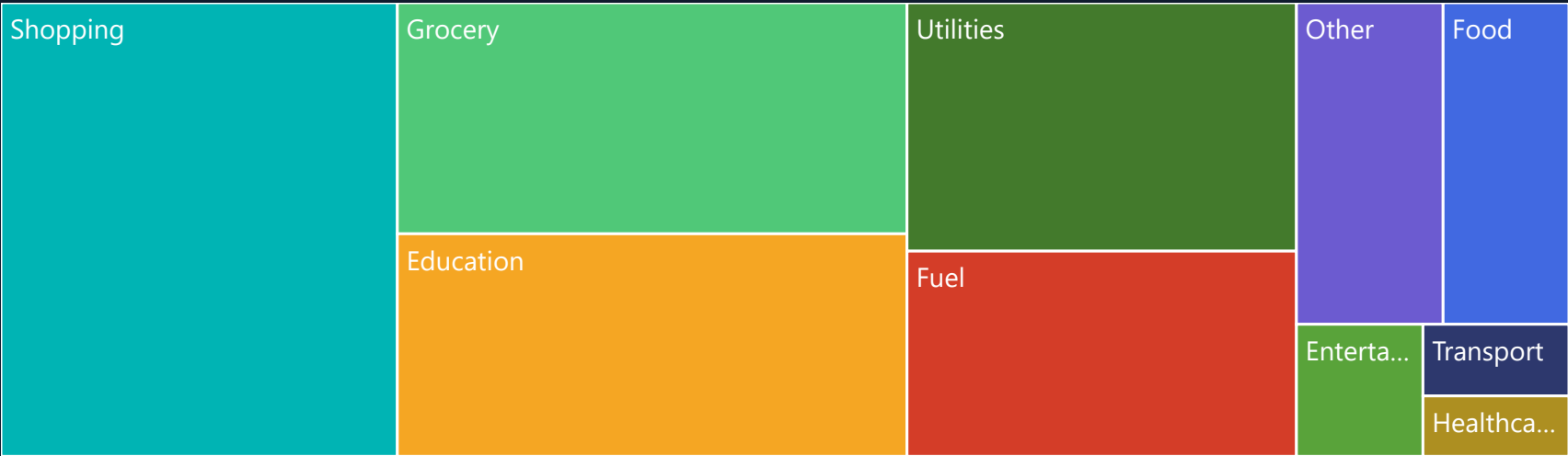


Fraud And Security Analysis

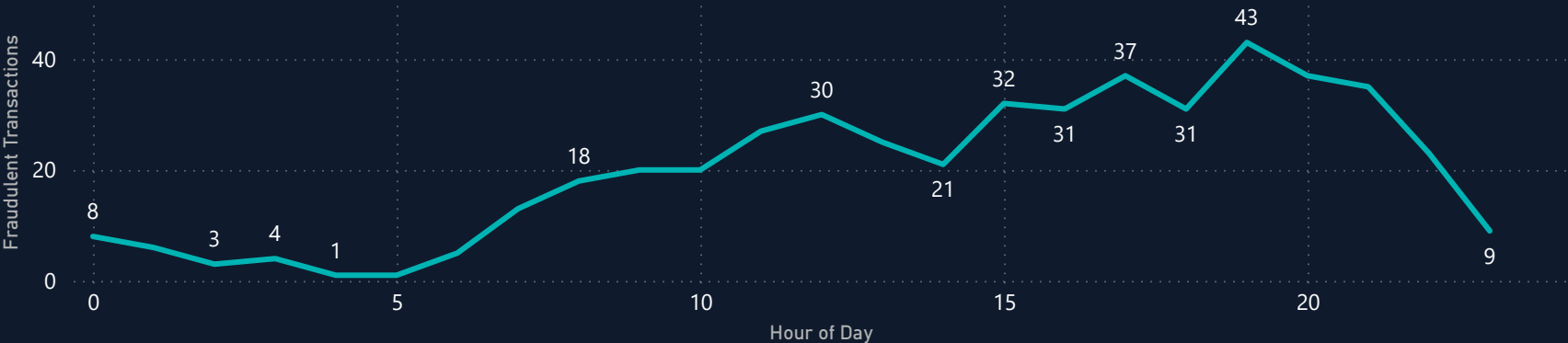
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- Model Explainability



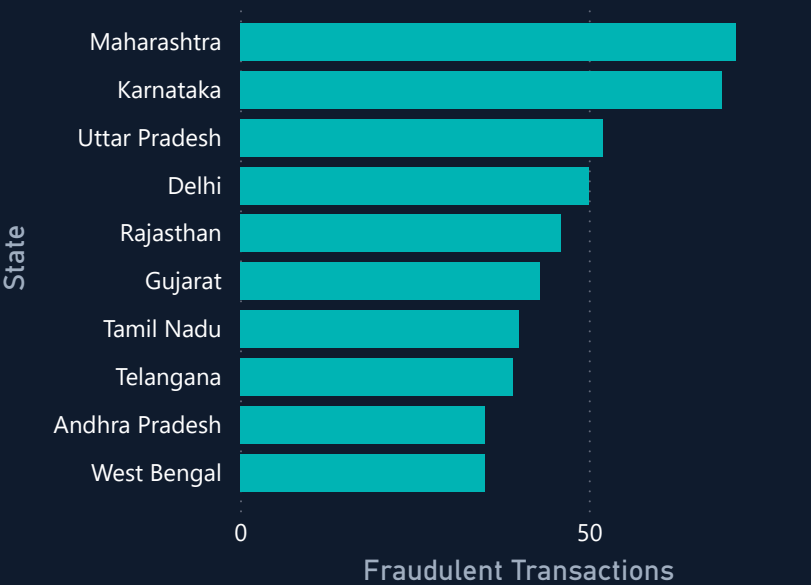
Shopping & Grocery Merchants Are Most Targeted



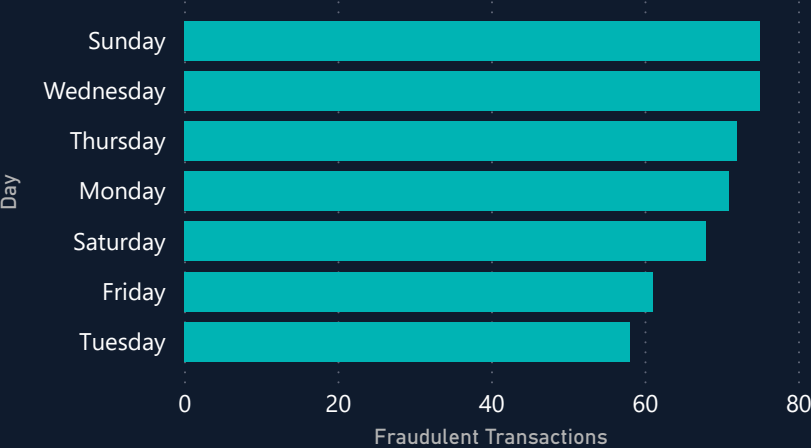
Fraud Spikes Sharply Between 8PM and 11PM



Which States Have the Most Fraudulent Transactions?



Which Days of the Week See the Most Fraud?



ML Model Performance

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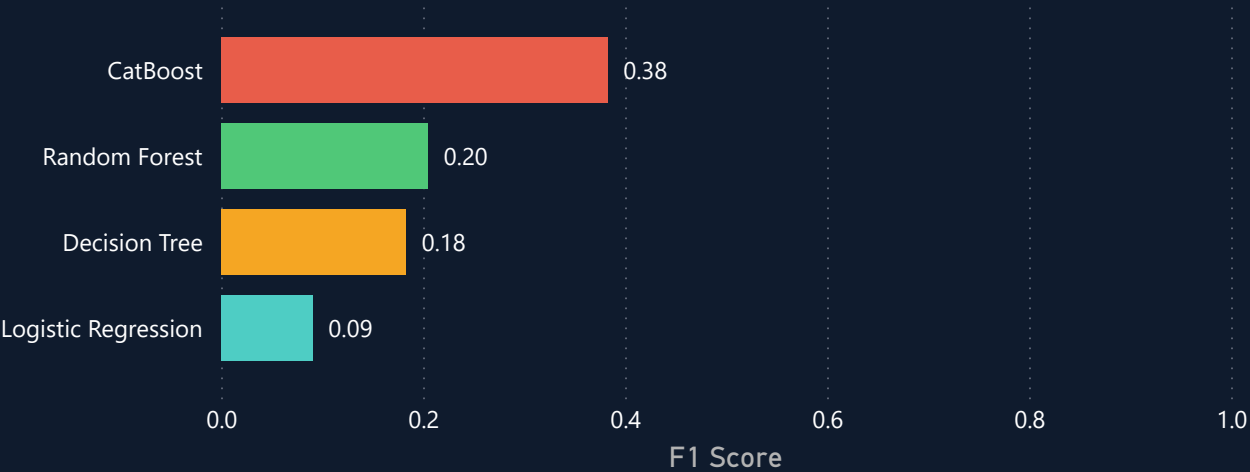
0.97
Best Accuracy

0.38
Best F1 Score

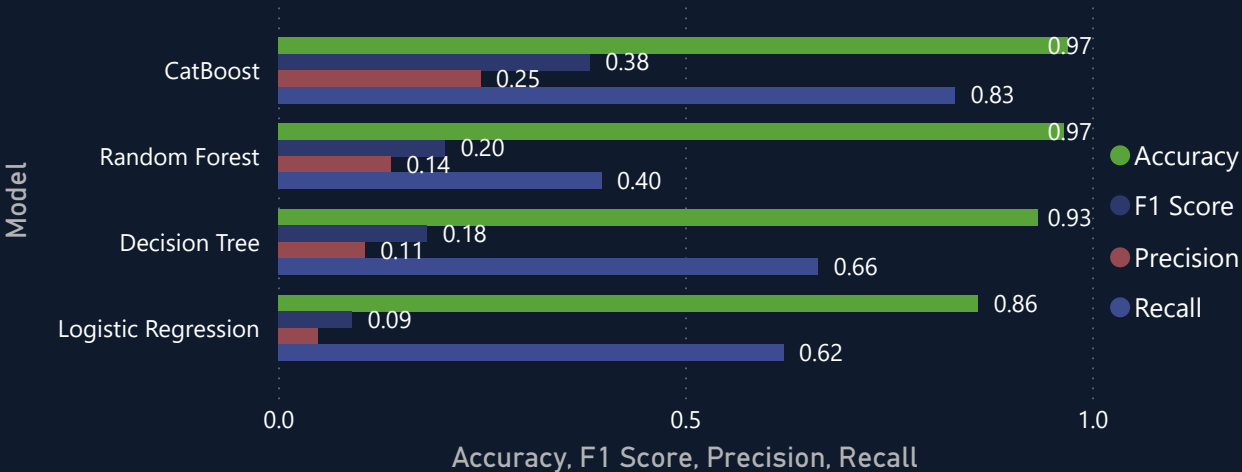
0.25
Best Precision

0.83
Best Recall

CatBoost Achieves the Best Fraud Detection F1 of 0.38



High Accuracy Alone Is Misleading — F1 Score Decides the Winner



How Do All 4 Models Compare Across Every Metric?

Model	Accuracy	F1 Score	Precision	Recall
Logistic Regression	0.86	0.09	0.05	0.62
Decision Tree	0.93	0.18	0.11	0.66
Random Forest	0.97	0.20	0.14	0.40
CatBoost	0.97	0.38	0.25	0.83

Best Model: CatBoost
Highest F1 Score — Best for Fraud Detection

Model Evaluation Curves

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0.92

CatBoost AUC

0.85

Decision Tree AUC

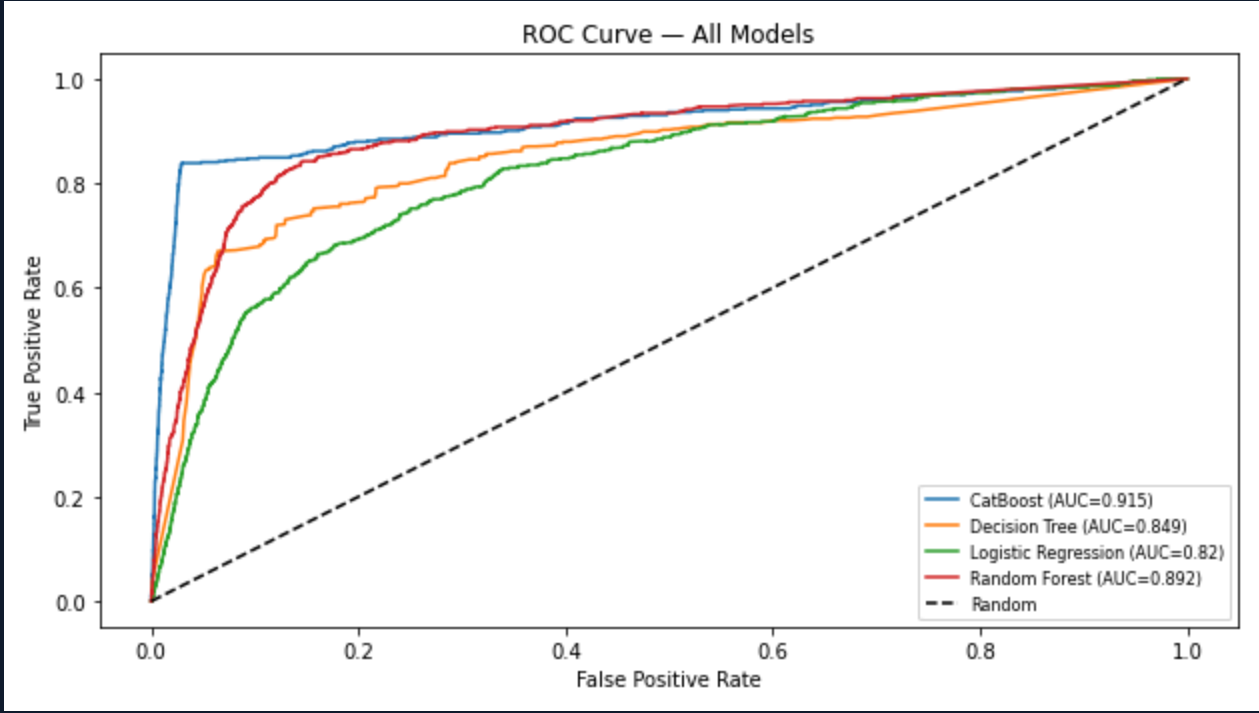
0.82

Logistic Regression AUC

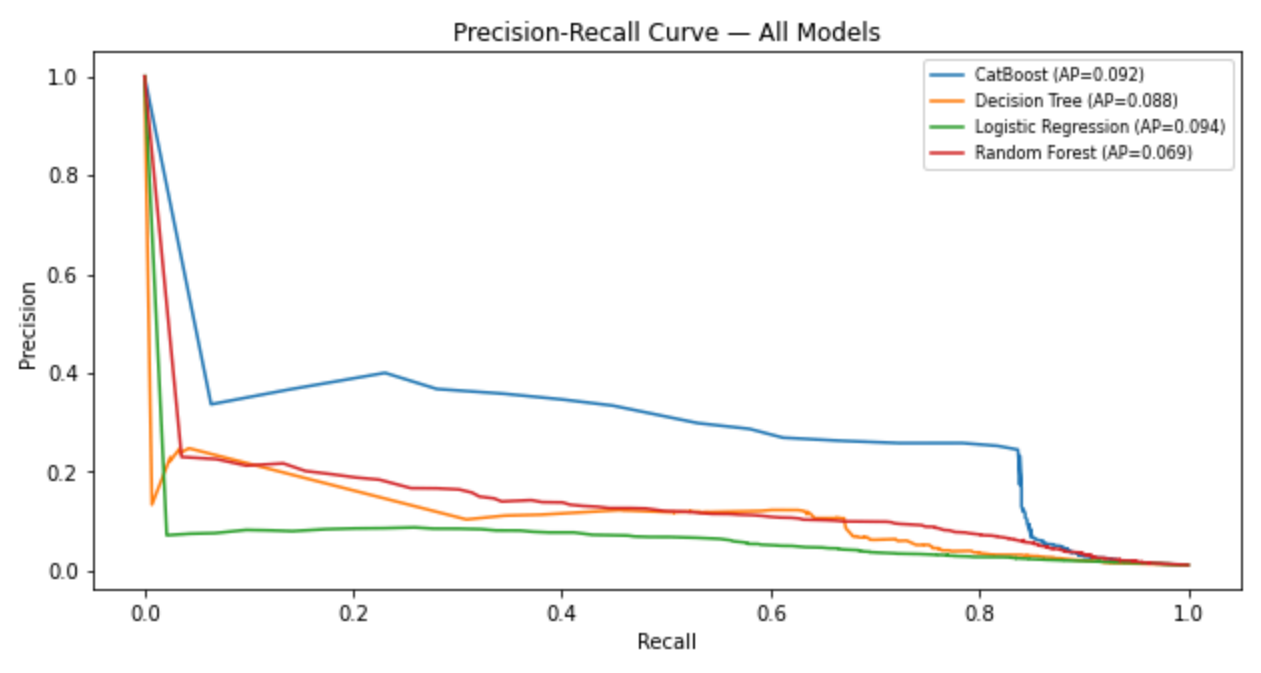
0.89

Random Forest AUC

ROC Curve — All Models



Precision-Recall Curve — All Models



Model Explainability

Threshold Tuning Chart

